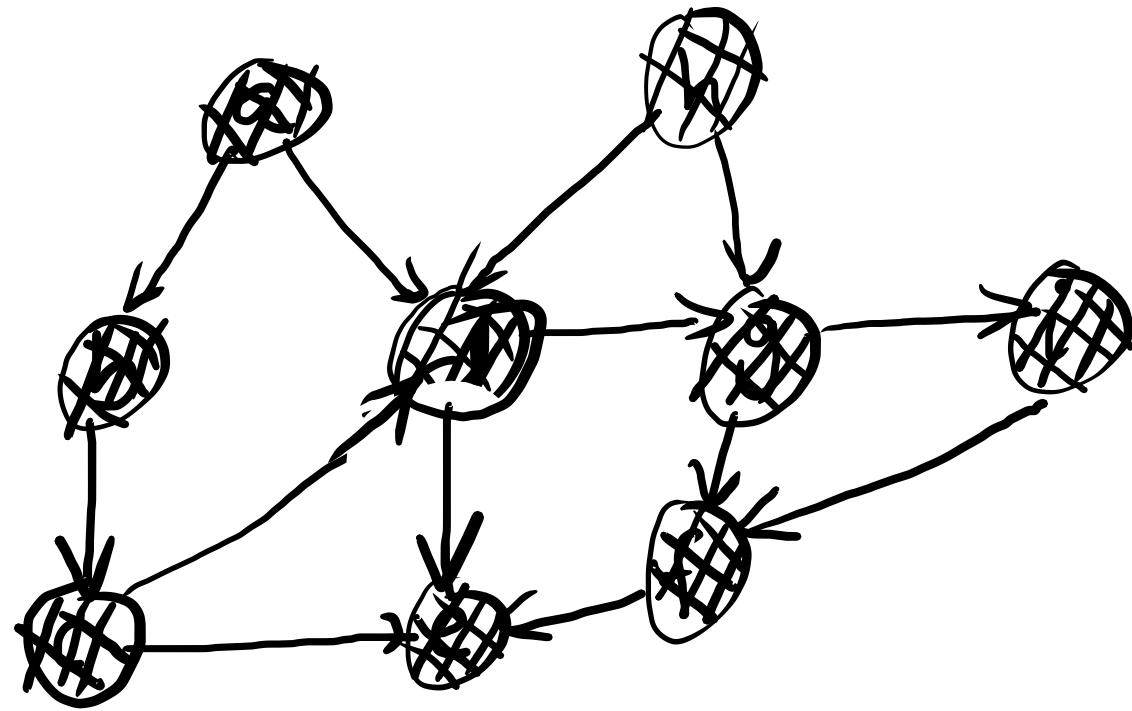


Cycle Detection using DFS



d
|
g
|
f
|
e

i

a
|
b
|
c

h
 $l(h) = [17, 18]$

forward edges
(d, g) (g, i) (g, f)
(f, e) (a, b) (b, c)

backward edges

(e, d)

cross edges

(i, f) (a, d) (c, d) (c, e)
(h, d) (h, g)

$$l(d) = [1, 10]$$

$$l(g) = [2, 9]$$

$$l(f) = [3, 6]$$

$$l(e) = [4, 5]$$

$$l(i) = [7, 8]$$

$$l(a) = [11, 16]$$

$$l(b) = [12, 15]$$

$$l(c) = [13, 14]$$

Time Complexity of DFS

$$O(|V| + |E|)$$

↑ ↑
for each vertex look at
 WHITE out-neighbors

Cycle Detection Time Complexity.

$$O(|V| + |E|) + |E|$$

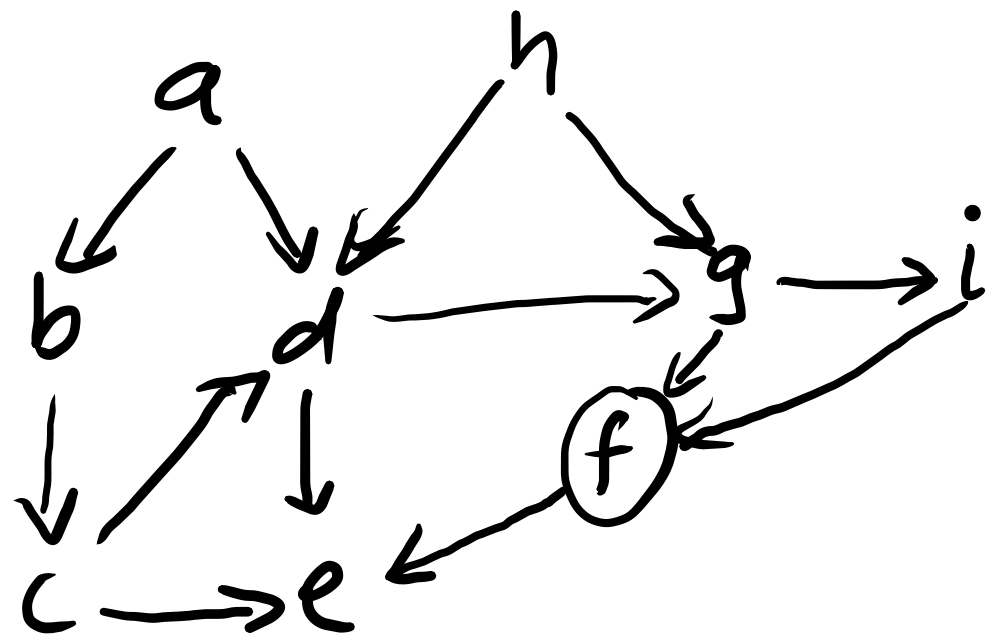
DFS ~~Determine~~
with time keeping type of edge

$$|V| + 2|E| \quad \text{OR} \quad O(|V| + |E|)$$

Topological order

Let $G = (V, E)$ be a directed acyclic graph (DAG)

A topological order of G is an ordering of the vertices V such that for any edge (u, v) it must hold that u precedes v in that order.



Two possible topological orders

h a b c d g i f e

a b c d g i f e h

Not a topological order

h a d b c g i f e

because (c, d) edge

DFS(G) $\rightarrow$ Timer updates

for each vertex $u \in G.V$

$u.color = WHITE$

$u.\pi = Nil$

time = 0

for each vertex $u \in G.V$

if $u.color == WHITE$

DFS-REC(G, u)

DFS-REC(G, u)

time = time + 1

$u.d = time$

$u.color = GRAY$

for each vertex $v \in Adj[u]$

if $v.color == WHITE$

$v.\pi = u$

DFS-REC(G, v)

$u.color = BLACK$

time = time + 1

$u.f = time$